🌻 ➕ ➖ Opposites

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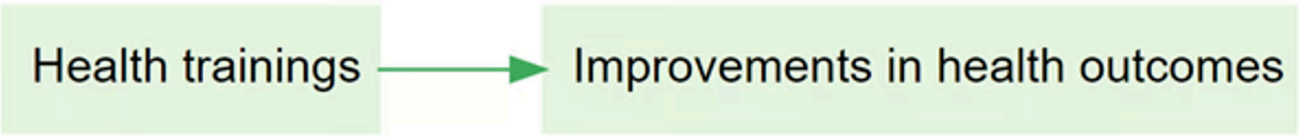
Introduction

In the first part of this Guide we have dealt only with undifferentiated links which simply say “C causally influences/influenced E” or more precisely “Source S claims/believes that C causally influences/influenced E.” We call this “barebones style” causal mapping. There is nothing more to this kind of causal map than links between factors. No other features are used.

Barebones-style mapping can be interpreted in two different ways:

- As in QuIP, to show causal influences between past events. It is an open question to what extent these causal claims can be generalised. It is possible to parse a link from C to E as saying not only that C is something which has the causal power to influence E but also that in some sense C happened and did in fact influence E, i.e. made a difference to it.
- To show only causal influences between factors, without recording what did or does happen.



In Part 1 of this Guide we did in fact also introduce an additional convention, hierarchical coding, in which the “;” separator is used to encode the idea that C; D can be read “D, an example of C” and that at a suitable level of abstraction we can approximate C; D as C.

Combining opposites

This section presents a simple and powerful way of dealing with “negative” factors such as `poor health` which are in some sense the opposite of existing “positive” factors such as `good health`.

In some kinds of causal mapping and systems diagramming, more sophisticated approaches are used in which the factors are considered to be variables and the links between them can have positive or negative strength. The approach we present here is a simpler alternative.

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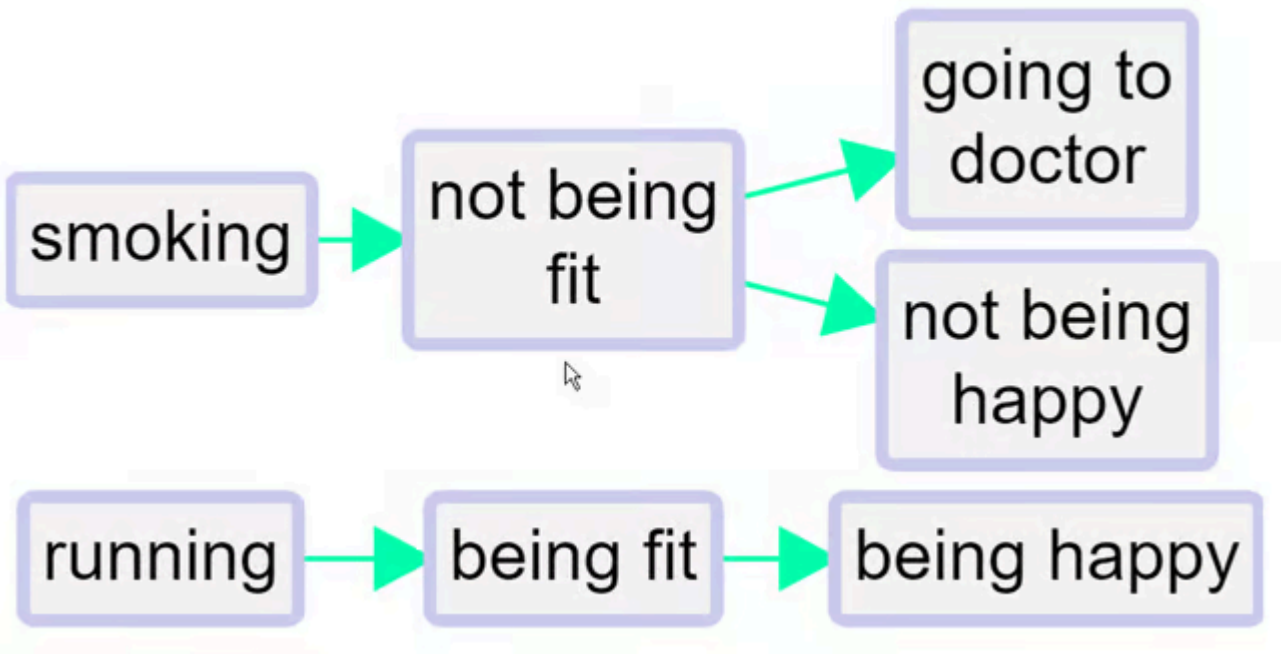
Here is an example of quite minimalist QuIP-style coding. There are the beginnings of some ideas about (and issues with) polarity: for example, we have `fit`and `not fit`.



We've all done this kind of coding, with classic examples being coding for both employment and unemployment or for both health and illness. This could for example be two different stories about two different people; or it could be different aspects of or periods within one person's life.

This minimalist style on its own is unsatisfactory. We haven't told the app that being fit is represented with a somehow positive and somehow negative factor. So can't join them up. We can't compare the way that being fit leads to happiness and on the other hand not being fit leads to unhappiness (and to visiting the doctor). **We can't for example deduce that running might make visits to the doctor less likely.** Also, if we produce a table or do other analyses focused on healthy habits, we might miss data on the closely related unhealthy habits.

The first step forwards is to follow this convention:

> To signal that two factors are opposites we formalise the idea we already instinctively used in the above example, where we used the word "not" for one of each pair. Formally, we will code them in the form "~Y" and "Y." The ~ may appear at the start of a factor label. This already ensures that when we search for "Y" we will also find "~ Y."In the Edit Multiple Factors panel, these two factors will be listed next to each other - the alphabetical listing will ignore the ~.

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We talk about *opposites* rather than positive/negative or plus/minus because that frees us from any implications about valence or sentiment: smoking is the opposite of not smoking, health is the opposite of not health / ill health / illness.

Where there is some kind of valence or sentiment involved, we do suggest using the ~ sign for the negative member of the pair. But it wouldn't make any difference to the app.

So the same map would look like this, using

```
~
```

instead of

```
not
```

.

image.png

Non-hierarchical coding with opposites is easy:

- Eating vegetables
- ~Eating vegetables
- Smoking
- ~Smoking

When you use the "combine opposites" filter (switched on),

Untitled

the app tries to combine any pairs of factors which are opposites. It looks at all the factors which begin with a ~ and takes off the ~ where there was one. But it only does this if there is in fact such an opposite already coded in the file as currently filtered, otherwise there wouldn't be any point.

image.png

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So now there are for example two factors combined into the “fit” factor and two into the “happy” factor. The “not” factors have their incoming and outgoing links preserved, but when a factor is flipped to match up with its opposite, the part of the link next to that factor are now coloured pink. So the lower link from fit to happy is pink because the factor at each end of the link has been flipped from “~Y” form to “Y” form; the influence factor was originally *not fit* and the consequence factor was originally *not happy*. So there is no danger of thinking that this is really just another case of the other link, i.e. of fitness leading to happiness.

So, a link has two polarities: a *from* polarity and a *to* polarity. If the signs of the two polarities are opposite, then the effect of the influence factor on the consequence factor is reversed.

Both links from fit to happy have the same overall polarity (normal, not reversed) but they do not represent the same information.

N**o information is lost when you press the “combine opposites” button; you can still always read off the original map from it.**

Opposites coding within a hierarchy

When using hierarchical coding, the sign “~” may appear at the start of a factor label *and/or at the start of any component in a factor label*.

Here is a similar story, now coded hierarchically. In this example, we only see ~ at the beginning of the factors, not yet within them.

image.png

When you press “combine opposites,” the app tries to combine any pairs of factors which are opposites. It looks at all the factors which begin with a ~ and *flips each component*, taking off the ~ where there was one, and inserting one where there was not. But again it only does this if there is in fact such an opposite already coded in the file as currently filtered, otherwise there wouldn’t be any point, because there is nothing to combine.

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Here is the same example, but also “zoomed out” to the top level.

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image.png

A quantitative social scientist might solve this problem by flipping the polarity of the negative examples, coding them as positive but using minus strengths for the connections. So smoking influences good health but with a minus strength. However this always seems somehow unsatisfactory and is complicated to do. It is particularly unsatisfactory when *both* ends of the arrow are flipped in this way so that we code the influence of being unfit on being unhappy as a green arrow from fitness to health!

By default in print view, links in the same bundle, i.e. with the same from and two factors, are no longer always displayed as one, with the frequency noted as a label. If we were using the quantitative approach, some of the links in the bundle may have minus rather than plus strength, etc, and we would have to somehow form some kind of average to arrive at an overall strength score, which is not at all satisfactory. Now, the links are only counted together if they have the same *from* and *to* polarities. So there can be up to four different links from one factor to another in Print view.

We are deliberately **not** falling Into the trap of somehow trying to aggregate the different *strengths* to say for example "there are 6 plus links from advocacy to compliance and 1 minus link so this is like 5 plus links because 6 - 1 = 5." We don't have evidence for an aggregated strength; we have aggregated evidence for a strength. Aggregating different pieces of *evidence* for links with different strengths is not the same as aggregating links with different strengths. So our more conservative approach preserves information.

It's also possible that someone says "I know this intervention works not only because the intervention made me happier but also because I saw the people who didn't get it and they are definitely not happier as a result." In this case, we might code both arrows, intervention ➜ happy and not intervention ➜ not happy.

Opposites coding within components of a hierarchy

Sometimes we need to use the ~ sign within the components of a hierarchy.

- ~Healthy habits; ~eating vegetables

is the opposite of

- Healthy habits; eating vegetables

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Not eating vegetables, which is an example of unhealthy behaviour, is the opposite of eating vegetables, an example of healthy behaviour.

- ~Healthy habits; smoking

is the opposite of

- Healthy habits; ~smoking

Smoking, which is an example of unhealthy behaviour, is the opposite of not smoking, an example of healthy behaviour.

So here we add one more causal claim to our above example, at the bottom:

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The healthy habit of not smoking leads to being fit.

So the app correctly detects that not smoking is the opposite of smoking:

image.png

The two arrows at bottom left (one all green, one all pink) show that there is one example of this particular healthy habit leading to fitness, and the complementary example in which the opposite of this habit leads to the opposite of fitness.

Zoomed out to the top level:

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Bivalent variables?

Note that this is the mutually exclusive condition from classical logic, but we don't have any mention of an exhaustive condition: for us, it is not the case that everything has to be either wealthy or poor.

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We could imagine that in a causal map, even before we even think about which specific factors might be opposites of other specific factors, each and every factor is really a kind of two-value factor like wealthy (as opposed to non-wealthy) or receiving tuition (as opposed to not receiving tuition). This is the quantitative way of thinking, in which every factor is really a variable which takes at least two values. But this is an unnecessary complication for us. The important point is that usually people don't think of the absence of something as having causal powers (though there are exceptions).

Suppose we are trying to code someone's understanding of income. They tell us that if when people are wealthy, they tend do certain things, and when they are poor they tend to do other things. To some extent, these two sets of things are themselves the reverse or contrary of one another. For example wealthy people will probably have the best education and poor people the least. Sometimes the sets of things we associate with the two opposite poles are not obviously a simple reflection of one another. For example, poor people are often hungry whereas non-poor people are not. But in an affluent country, not being hungry is probably not an important feature we would think of to describe being wealthy as opposed to not-wealthy, if we can assume that non-wealthy people are *overall* rarely hungry. We are more likely to think of things like eating often in posh restaurants.

So being wealthy and being poor are a good example of factors which we *should* consider coding as opposites. We can call the resulting factor, when we press the combine opposites button, *bivalent*.

, we could imagine there was a middle point too:

- Wealthy
- Not-wealthy / Not-poor - a kind of midpoint
- Poor

This is the usual case.

Which pairs of factors should we consider for opposites coding?

Short answer: use opposites coding for a pair of factors X and Y (i.e. recode Y as ~X or recode X as ~Y) if both X and Y naturally occur, separately, in the coding, but they can be considered, broadly speaking, as opposites of one another:

- in particular, it wouldn't normally make sense to apply both of them at the same time in the same situation (you can't be both wealthy and poor in the same sense at the same time)

If in doubt about which of the pair to recode, we usually pick X as the primary member of the pair if it is:

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- usually considered as positive / beneficial / valuable
- and/or usually associated with “more” of something rather than “less” of something.

⚒️ Colour of factor borders (not currently implemented)

But there is more: the border colour could reflect the overall polarity of the factor.

One possibility is: The more *plus* incoming and outgoing arrows there are, the greener it is, and the more *minus* arrows there are, the redder it is. If the balance is equal, the border is light grey. If a factor has a red border, that means that at least mostly, its *opposite* was mentioned. So in this example, happy has a grey border because it was mentioned once and its opposite was mentioned once. Fit has a slightly pink border because it has one incoming and one outgoing plusses, and one incoming and two outgoing minuses.

However this reflects information which is actually already visible in the diagram. A second possibility is to use the same green-red scale to reflect the proportion of factors contributing to a factor which have been flipped: this information can not otherwise be deduced from the map. It becomes particularly relevant when coding opposites within a hierarchy, see below. However this colour scheme might be misinterpreted, as it is only affected by the number and polarity of factors which have been collapsed and not by the number of links.

The total of the numbers on the arrows (here all the numbers are 1 so they are not shown) into and out of a factor is its citation count. But now we have additional information and its citation count is also equal to the number of times it was mentioned in plus form and the number of times it was mentioned in the opposite form.

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